Oxygenation versus tissue composition in positronium-lifetime PET: an identifiability limit for hypoxia imaging

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Author: Ernest Darell Zermeño

Affiliations

1. Universidad Panamericana, Mexico City, Mexico.
2. Universidad de Guadalajara, Guadalajara, Mexico.

Corresponding author: Ernest Darell Zermeño, 0244552@up.edu.mx

Abstract

Purpose. Positronium lifetime imaging with long-axial-field-of-view PET/CT has been proposed as a biomarker of tissue oxygenation. The same lifetime also depends on tissue composition. We quantified whether oxygenation and composition can be separated from positronium observables, and what bias arises if composition is ignored.

Methods. A rate-additivity model mapped pO_2 and lipid/free-volume fraction to the ortho-positronium lifetime τ_3 and intensity I_3. We derived the apparent-pO_2 bias of a water-calibrated reader, analysed Fisher-information rank for τ_3 alone, computed Cramér-Rao bounds (CRLBs) for (pO_2, f) from a Poisson lifetime-spectrum model with nuisance profiling, and translated the bounds into counts and region-of-interest volumes.

Results. From τ_3 alone, oxygenation and composition were structurally non-identifiable because both enter one summed decay rate. Ignoring composition produced apparent-pO_2 errors of -178, -439 and -860 mmHg at 2%, 5% and 10% lipid for tissue at 20 mmHg. Adding I_3 restored identifiability in principle when the (λ_3, I_3) Jacobian was non-collinear; in the pessimistic baseline $\partial I_3/\partial pO_2 \approx 0$, this required $\partial I_3/\partial f \neq 0$. The pO_2 CRLB remained 330-760 mmHg at 10^5 o-Ps counts. A 10 mmHg target required litre-scale pooling in current yield scenarios and was unreachable at native 4 mm voxel scale. Under an optimistic high-lipid-quenching stress test, small fatty ROIs approached feasibility, but lean tumour-like tissue remained litre-scale.

Conclusion. Positronium-lifetime hypoxia imaging is limited by an oxygenation-versus-composition identifiability problem. Lifetime-only maps should not be interpreted as oxygenation maps without composition control and a second observable.

Keywords: positronium lifetime imaging; PET/CT; hypoxia; oxygenation; identifiability; Cramér-Rao bound

Introduction

Ortho-positronium (o-Ps), the triplet bound state formed in a fraction of positron-electron annihilations, localises in nanometre-scale free-volume regions and annihilates in matter predominantly through pick-off processes. Its lifetime is therefore sensitive to tissue microstructure and composition. Molecular oxygen provides an additional paramagnetic quenching channel through spin conversion, making the o-Ps lifetime an attractive candidate biomarker for oxygenation and tumour hypoxia [1, 2]. This is clinically relevant because hypoxic tumour regions are more resistant to radiotherapy and systemic therapy, while established nuclear-medicine hypoxia tracers have practical limitations.

The technical basis for in-vivo positronium lifetime imaging (PLI) has advanced rapidly. Dedicated multi-photon systems and long-axial-field-of-view PET/CT scanners now support in-vivo positronium measurements and regional lifetime estimates [3-6]. These developments make the oxygenation question timely. However, the o-Ps lifetime is not oxygen-specific. Across soft tissues, composition-driven lifetime variation is on the order of hundreds of picoseconds, whereas the oxygen-specific term implied by aqueous quenching calibrations is on the order of picoseconds across physiological pO_2 differences [1, 7]. Calibrations performed in water or aqueous samples control composition by design and therefore do not test the in-vivo confound.

The relevant prior work has established the ingredients separately. Shibuya and Stepanov established oxygen quenching and its spin-conversion mechanism [1, 2]; Avachat et al. showed that lifetime separates soft tissues by composition/free-volume [7]; Moskal and Mercolli demonstrated in-vivo PLI and LAFOV feasibility [3-6]; photon-economy studies address estimation of a single lifetime [8, 9]; and $3\gamma/2\gamma$ oxygenation work motivates multi-observable readouts [10-12]. What has been missing is a quantitative statement of whether oxygenation and composition are identifiable as separate variables, how large the apparent hypoxia bias can be, and how many counts would be needed to overcome the degeneracy with a second observable.

Here we formulate the problem as an identifiability and CRLB analysis for an emerging nuclear-medicine biomarker. We quantify the bias produced by a lifetime-only, water-calibrated interpretation; prove the structural rank defect of the lifetime-only inverse problem; evaluate whether adding o-Ps intensity I_3 or a related $3\gamma/2\gamma$ observable can break the degeneracy; and translate the resulting precision limits into count and volume requirements. A related single-subject cardiac PLI re-analysis reported an in-vivo lifetime contrast not explained by blood oxygenation, illustrating why the general identifiability problem matters in human data [13].

Material and methods

Forward model

We modelled the o-Ps decay rate as a sum of independent channels:

$$\lambda_3 = 1/\tau_3 = \lambda_{\text{pickoff}}(f) + \kappa(f)\alpha(f)pO_2,$$

where f is an effective lipid/free-volume-rich composition coordinate in a two-compartment tissue, $\lambda_{\text{pickoff}}(f)$ is the composition-dependent baseline rate, κ is the oxygen quenching constant and α is oxygen solubility. The baseline lifetime $\tau_0(f)$ interpolated between water and lipid endpoints. The water endpoint was used as an operational PET/LAFOV water-reference lifetime, anchored to the 1.815 +/- 0.013 ns water measurement reported by Steinberger et al. [14] and consistent with the approximately 1.82 ns water-rich endpoint measured with ^{124}I PLI by Mercolli et al. [15], rather than as a universal deoxygenated pick-off constant. The lipid:water oxygen-solubility ratio was anchored to published oil/lipid measurements [16, 17]. The o-Ps intensity was modelled as

$$I_3 = I_{3,w} + a_f f + a_p pO_2,$$

where $a_f = \partial I_3/\partial f$ and $a_p = \partial I_3/\partial pO_2$. The two load-bearing unknowns were made explicit: κ_{lipid}, swept over 0.1-1.0 times the water value as a conservative baseline sweep, and the composition slope $\partial I_3/\partial f$ (Table 1). The baseline scenario assumed the pessimistic case $\partial I_3/\partial pO_2 \approx 0$, so that oxygen affects the lifetime rate but not o-Ps formation probability.

Apparent-pO_2 bias

We defined a naive reader as an estimator that assumes pure-water calibration ($f = 0$) and inverts an observed lifetime to pO_2. The resulting apparent-pO_2 error was computed over true pO_2 and lipid fraction. This bias quantifies the oxygenation error that would arise when a composition-dependent lifetime is interpreted as oxygen-specific.

Identifiability and CRLB analysis

For $\theta = (pO_2, f)$, Fisher information matrices were computed from τ_3 alone and from the two-observable vector (τ_3, I_3). Structural non-identifiability was defined by rank deficiency of the Fisher information matrix; practical identifiability was assessed by the CRLB and parameter correlation. From τ_3 alone, both parameters enter through the scalar rate λ_3. With the single score row

$$J = [\, \partial\lambda_3/\partial pO_2,\ \partial\lambda_3/\partial f\,],$$

the lifetime-only Fisher information is

$$F_\tau = \sigma_\lambda^{-2} J^\top J, \qquad \det F_\tau = 0.$$

Therefore the Fisher information has rank at most 1 and the individual CRLBs for pO_2 and f diverge. With (τ_3, I_3), let $a = \partial\lambda_3/\partial pO_2$, $b = \partial\lambda_3/\partial f$, $p = \partial I_3/\partial pO_2$ and $c = \partial I_3/\partial f$. The two-observable problem is locally identifiable only when

$$\det J_{\lambda,I} = a\,c - b\,p \neq 0,$$

which is the exact non-collinearity condition. In the pessimistic baseline $p \approx 0$, this reduces to $c = \partial I_3/\partial f \neq 0$ because $a \neq 0$. This CRLB/identifiability framing follows the same logic used in other multi-parameter imaging inverse problems [19].

Poisson spectrum and nuisance profiling

To avoid treating τ_3 and I_3 as independent clean measurements, we built a binned Poisson lifetime-spectrum model. The expected spectrum contained a fast exponentially modified Gaussian component, an o-Ps component with slope $\tau_3(\theta)$ and amplitude $I_3(\theta)$, and a large flat accidental background. The per-bin Fisher information was

$$F_{ab} = \sum_k m_k^{-1}\,\partial_a m_k\,\partial_b m_k,$$

where m_k is the expected count in bin k. This includes the τ_3-I_3 estimation covariance. We also profiled spectral nuisance parameters by extending the Fisher information matrix to include background level, fast-component lifetime, prompt offset and instrument resolution, then marginalising to (pO_2, f). As an internal estimator check, Poisson spectra were simulated under the same forward model and fitted by maximum likelihood to verify attainability of the fixed-nuisance CRLB (Online Resource 1).

Counts and volume model

For the count-to-volume translation, we used a scenario anchor from a published ^{44}Sc PLI phantom study: approximately 3.6×10^4 o-Ps counts in a 5.57 mL region [18]. We scaled o-Ps counts by region-of-interest volume and by scenario-level effective isotope-yield factors representing prompt branching and source-to-background conditions. The scenarios were ^{124}I (best yield), ^{68}Ga/^{44}Sc (reference) and ^{82}Rb (lower yield). These are planning scenarios, not universal scanner constants. For each tissue state, we computed the region volume required to reach $\sigma(pO_2) \leq 10$ mmHg and classified a virtual tumour at 4 mm voxel scale as viable, marginal or not achievable.

All analyses were implemented in Python 3.13 using NumPy, SciPy and Matplotlib. Code, constants and generated results are available in the public repository listed in Data Availability. Use of AI-assisted tools is reported in Statements and Declarations.

Results

Lifetime-only interpretation produces large apparent oxygenation errors

The composition-dependent lifetime shift dominated the oxygen-specific lifetime term. For tissue truly at $pO_2 = 20$ mmHg, a water-calibrated lifetime read produced apparent-pO_2 errors of -178 mmHg at 2% lipid, -439 mmHg at 5% lipid and -860 mmHg at 10% lipid (Fig. 1). These values are physiologically implausible, indicating that the uncorrected lifetime read is composition-dominated rather than oxygen-specific. The result is conditional on the endpoint-mixing forward model, but not fragile: the bias scaled approximately linearly with the composition-to-lifetime coupling in sensitivity tests, and fell below 10 mmHg at 5% lipid only if that coupling was approximately 44-fold smaller than the measured water-adipose baseline gap.

The oxygen sensitivity was also composition-dependent, temperature-dependent and uncertain. At body temperature the aqueous sensitivity was approximately 0.09 ps/mmHg (O_2 solubility approximately 1.3 microM/mmHg at 37 degrees C), whereas the aqueous calibration of Shibuya et al. near room temperature (approximately 20 degrees C) implies approximately 0.13-0.14 ps/mmHg; the difference reflects the temperature dependence of O_2 solubility, and the smaller body-temperature value is the one relevant in vivo. Because κ_{lipid} has not been measured in relevant lipid/protein environments, the conservative baseline model predicted a net range of approximately 0.09-0.90 ps/mmHg in lipid-rich material across the κ_{lipid} sweep (Fig. 2).

τ_3 alone is structurally non-identifiable

From the lifetime alone, pO_2 and composition were not separately identifiable. Both parameters enter only through the single summed decay rate λ_3, so the two-parameter Fisher information matrix has rank 1 and the individual CRLBs diverge. This is an algebraic non-identifiability, not a shortage of counts.

Adding I_3 restored identifiability in principle, because composition moves both τ_3 and I_3 while, in the pessimistic baseline model, oxygen moves only the rate. The exact local condition is $\det J_{\lambda,I} = a\,c - b\,p \neq 0$ (Fig. 3). In the baseline $p = \partial I_3/\partial pO_2 \approx 0$, the condition reduces to $\partial I_3/\partial f \neq 0$; if that slope approaches zero, the system again becomes degenerate. The dependence on the composition slope was large even in the fixed-nuisance Poisson-spectrum model (Table 2).

The second-observable rescue is statistically weak

At 10^5 o-Ps counts and a representative hypoxic fatty state (pO_2 = 20 mmHg, $f = 0.10$), the pO_2 CRLB remained large across noise models: 332 mmHg for the fixed-nuisance Poisson spectrum, 570 mmHg for an idealised two-observable model using empirical lifetime precision from the field, 665 mmHg after profiling background and fast-component nuisances, and 758 mmHg after profiling all tested nuisances (Table 3). The spectral model also exposed substantial τ_3-I_3 covariance, with parameter correlation $\rho = 0.70$ in the optimistic Poisson model compared with $\rho = 0.05$ in the idealised two-observable model.

The count scaling implied approximately 3-5 × 10^8 o-Ps events for a 10 mmHg target in the profiled models, depending on the noise model (Fig. 4). The spectrum calculations are expressed in total in-window events, with $N_3 = I_3(1 - \text{background})N_{\text{total}}$; at the representative state used here, 10^5 o-Ps counts correspond to approximately 1.1×10^6 in-window events. Parametric Monte Carlo confirmed that the fixed-nuisance bound is statistically attainable under the assumed spectrum model: empirical standard deviations matched the CRLB at both tested count levels, and the pO_2 pull distribution had mean +0.05 and standard deviation 1.00 (Online Resource 1). This validates estimator behaviour under the specified model; it does not validate the unmeasured biological constants.

Voxel-scale pO_2 separation is not achievable under current yield assumptions

When translated to region-of-interest volumes, clinically useful separation required litre-scale pooling. At pO_2 = 10 mmHg and 5% lipid, the required pooled volumes were approximately 20 L for the best-yield ^{124}I scenario, 78 L for ^{68}Ga and 130 L for ^{82}Rb (Fig. 5). At 60% lipid, where oxygen sensitivity is higher, the best-yield scenario still required approximately 5 L. At a native 4 mm clinical voxel (0.064 mL), pO_2 separation was not achievable anywhere in the virtual tumour phantom (Fig. 6).

Because κ_{lipid} is unmeasured, we also performed an intentionally high stress test above the conservative baseline sweep. Increasing κ_{lipid} markedly improved fatty-tissue feasibility, but did not remove the structural non-identifiability or make the lean-tissue tumour-hypoxia regime voxel-feasible. In the ^{124}I scenario, the required volume at $f = 0.60$ and pO_2 = 10 mmHg fell from 5.1 L at the central 0.3× assumption to 98 mL at 3× and 9.5 mL at a 10× organic-liquid stress test. At $f = 0.05$, however, the same stress test still required litre-scale pooling: 19.6 L, 7.1 L and 1.7 L at 0.3×, 3× and 10×, respectively (Table 4). Thus a small fatty ROI could become conceivable under an optimistic organic-liquid-like quenching stress test, but lean tumour-hypoxia tissue remains litre-scale and the native-voxel conclusion and need for a second observable are unchanged.

Discussion

This analysis identifies a fundamental limitation for positronium-lifetime hypoxia imaging. Oxygen quenching of o-Ps is real, but a lifetime map is not an oxygenation map unless composition is controlled. From τ_3 alone, oxygenation and composition collapse into a single measured decay rate, making the inverse problem structurally non-identifiable even with arbitrarily many counts. The practical consequence is large apparent-pO_2 bias when composition is ignored.

The analysis is also constructive. It shows that a second observable, such as o-Ps intensity I_3 or a related $3\gamma/2\gamma$ ratio, is required to break the degeneracy. This is the o-Ps analogue of adding a second echo to make fat/water separation full-rank: one contrast dimension is not enough when two physical sources move the signal. In the pessimistic baseline model, the key separability handle is the composition dependence of I_3. Therefore, PLI oxygenation studies should report the lifetime together with I_3 or $3\gamma/2\gamma$, and should treat co-registered CT/MRI-derived composition as a correction variable rather than a nuisance to ignore.

However, identifiability in principle does not imply clinical feasibility. The CRLBs show that current count yields are far from sufficient for voxel-level pO_2 separation. The relevant design target is not merely estimating a single lifetime, for which photon-limited precision scaling is well established [8, 9], but separating oxygenation from composition in a two-parameter model. Under the yield scenarios used here, separation is out of reach at native voxel resolution by approximately five to six orders of magnitude. The smaller three-to-four-order gap applies only after pooling to millilitre-scale regions. This does not preclude regional, ex-vivo or future high-yield studies, but it sets a quantitative bar for claims of voxel-scale hypoxia imaging.

The decisive experiment is a controlled PALS matrix over O_2 by medium: water, lipid emulsion and protein/serum. Such an experiment should measure both τ_3 and I_3 under independently measured dissolved oxygen. It would determine κ_{lipid}, protein-environment quenching, $\partial I_3/\partial f$ and $\partial I_3/\partial pO_2$, converting the present conditional model into a measured one. This is a modest bench experiment compared with a full clinical imaging study and should precede strong clinical claims about PLI hypoxia.

This study has limitations. The forward model is intentionally transparent and two-compartment; it does not include richer Tao-Eldrup free-volume physics or chemical effects such as inhibition and radical scavenging. The I_3 values are illustrative placeholders, and the composition slope is assumed non-zero in the baseline scenario. The lipid quenching constant is unmeasured in relevant biological environments and was therefore swept. The count model is anchored to one published phantom-study yield and uses scenario-level isotope factors rather than scanner-specific measured yields. Nuisance profiling provides a more conservative CRLB, but the spectrum model remains idealised. In particular, I_3 in the model is the intrinsic o-Ps formation probability, whereas a scanner estimates a fitted o-Ps amplitude that also absorbs detection efficiency, scatter, attenuation, geometric acceptance, accidental background and prompt-gamma contamination. The FIM profiles per-spectrum nuisances, but not this full imaging chain; the realistic I_3 covariance and the corresponding CRLB are therefore likely lower bounds on the in-system problem. Finally, the Monte Carlo validates statistical attainability under the assumed spectrum model, not the biological forward model. A PALS measurement or re-analysis of real data with I_3 extraction remains the key empirical anchor.

In summary, positronium-lifetime hypoxia imaging is limited by an oxygenation-versus-composition identifiability problem. Lifetime-only PLI contrasts should not be interpreted as hypoxia without composition control. A second observable can make the problem identifiable, but current count yields imply that clinically useful voxel-scale separation is not achievable without major gains in effective o-Ps statistics or strong external composition constraints. Highly fatty tissue may be less prohibitive under an optimistic high-κ_{lipid} stress test, but that does not rescue the lean tumour-hypoxia regime targeted by voxel-scale hypoxia imaging claims.

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Statements and Declarations

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Author contributions. E.D.Z. conceived the study, implemented the model, performed the analysis, generated the figures and wrote the manuscript.

Data availability. No clinical or animal datasets were analysed. The datasets generated during the current study, including code, constants, intermediate outputs and figures, are available at https://github.com/Jefemaestro33/oxygenation-versus-tissue-composition-in-positronium-lifetime-pet-identifiability-limit.

Code availability. The analysis code is available at https://github.com/Jefemaestro33/oxygenation-versus-tissue-composition-in-positronium-lifetime-pet-identifiability-limit under the repository licence terms.

Ethics approval. Not applicable. This is a modelling and methods study using no human participants, human data, animals or animal data.

Consent to participate. Not applicable.

Consent to publish. Not applicable.

Use of AI tools. AI-assisted programming and editorial tools were used to help cross-check code, draft text and identify consistency issues under the author's direction. The author reviewed all outputs and takes responsibility for the final manuscript, analysis and interpretation.

Tables

Table 1 Forward-model constants and load-bearing unknowns

Quantity	Value	Status	Source
τ_0 water	1.815 ns	operational water reference	[14, 15]
τ_0 adipose	2.5-2.7 ns	measured	[7]
$\kappa(O_2)$ water	0.0204 µmol^{-1} µs^{-1} L	measured	[1, 2]
O_2 solubility lipid:water	approximately 5×	measured	[16, 17]
Quenching mechanism	spin conversion	measured	[2]
κ_{lipid}	unmeasured; conservative baseline sweep 0.1-1.0 times water	unknown	proposed experiment
$\partial I_3/\partial f$	unmeasured; assumed non-zero in baseline scenario	unknown	proposed experiment

Table 2 Dependence of the pO_2 CRLB on the composition sensitivity of I_3 at $pO_2 = 20$ mmHg, $f = 0.10$ and 10^5 o-Ps counts in the fixed-nuisance Poisson-spectrum model. I_3 is a dimensionless formation-probability fraction with $I_{3,w} = 0.15$ in the nominal placeholder model.

$\partial I_3/\partial f$	$\sigma(pO_2)$ (mmHg)
0.150 (nominal)	332
0.050	473
0.015	998
0	degenerate

Table 3 Cramér-Rao bound on pO_2 at $pO_2 = 20$ mmHg, $f = 0.10$ and 10^5 o-Ps counts

Noise model	$\sigma(pO_2)$ (mmHg)	ρ
Poisson, fixed nuisances	332	+0.70
Idealised two-observable	570	+0.05
Poisson, background+fast profiled	665	–
Poisson, all nuisances profiled	758	–

Table 4 Stress test of the unmeasured lipid quenching constant in the ^{124}I effective-yield scenario at $pO_2 = 10$ mmHg. Values are ROI volumes needed for $\sigma(pO_2) \leq 10$ mmHg.

$\kappa_{\text{lipid}}/\kappa_{\text{water}}$	$f = 0.60$	$f = 0.05$
0.3 (central baseline)	5.1 L	19.6 L
1.0 (water-value baseline)	0.75 L	14.3 L
3.0 (high stress test)	98 mL	7.1 L
10 (organic-liquid stress test)	9.5 mL	1.7 L

Figure legends

Fig. 1 Apparent pO_2 error produced by a water-calibrated lifetime interpretation when tissue composition is ignored. The contour marks the region where absolute error is 10 mmHg. Even low lipid fractions drive errors of hundreds of mmHg

Fig. 2 Oxygen sensitivity of the o-Ps lifetime as a function of lipid fraction. The band represents the conservative baseline sweep over the unmeasured lipid quenching constant κ_{lipid} from 0.1 to 1.0 times the water value; dashed/dotted curves show the 3× and 10× high-stress cases used in Table 4.

Fig. 3 Separability geometry of (τ_3, I_3). Iso-τ_3 contours show the near-degenerate lifetime direction, while iso-I_3 contours provide the composition handle. The shallow crossing reflects identifiability in principle but weak statistical separation

Fig. 4 CRLB $\sigma(pO_2)$ versus total in-window events for the fixed-nuisance full-spectrum (τ_3, I_3) fit. The lifetime-only case is structurally non-identifiable. A 10 mmHg target requires order-10^9 in-window events, corresponding to order-10^8 o-Ps counts after the $N_3 = I_3(1 - \text{background})N_{\text{total}}$ conversion.

Fig. 5 Region-of-interest volume required to separate pO_2 from composition with $\sigma(pO_2) \leq 10$ mmHg for three effective isotope-yield scenarios. Clinical separation requires litre-scale pooling under current assumptions

Fig. 6 Virtual tumour simulation. The left panel shows true pO_2; the right panel shows voxel-level separability at 4 mm resolution under the profiled CRLB model. pO_2 separation is not achievable anywhere at native voxel scale

Supplementary information

Online Resource 1 Parametric Monte Carlo validation of the fixed-nuisance Poisson CRLB under the assumed spectrum model. Pulls of $\hat{p}O_2$ are close to $N(0, 1)$, showing that the maximum-likelihood estimator reaches the fixed-nuisance CRLB when the forward model is correctly specified.